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Oefeningenreeks 1E nr. 34

Zoek een derde graad functie $A(z)$ met $\{1, -1\}$ als nulpunten.

$A(0) = 2 + i$ en $A(i) = 4$

$A(z) = az^3 + bz^2 + cz + R$

$A(0) = R = 2 + i$

$A(z) = az^3 + bz^2 + cz + 2 + i$

Vul 1 en -1 in $\Rightarrow \left\{ \begin{array}{l} a + b + c + 2 + i = 0 \\ -c + b - c + 2 + i = 0 \end{array} \right| \begin{array}{l} 1 \\ 1 \end{array}$

$2b + 4 + 2i = 0$

$b = -2 - i$

Vul -1 en i in $\Rightarrow \left\{ \begin{array}{l} -a + b - c + 2 + i = 0 \\ -ai - b + ci + 2 + i = 4 \end{array} \right| \begin{array}{l} 1 \\ 1 \end{array}$

$-a - c - ai + ci = -2i$

$\left\{ \begin{array}{l} -a - c = 0 \\ -ai + ci = -2i \end{array} \right.$

$\left\{ \begin{array}{l} a = -c \\ ci + ci = -2i \end{array} \right.$

$\left\{ \begin{array}{l} a = -c \\ 2ci = -2i \end{array} \right. \Rightarrow a = 1 \text{ en } c = -1$

Eindresultaat

$A(z) = z^3 + (-2 - i)z^2 - z + 2 + i$

References